

泰勒：

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \dots$$

不定积分：

$$a^n + b^n = (a+b)(a^{n-1} - a^{n-2}b + a^{n-3}b^2 - \dots + b^{n-1})$$

$$a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \dots + ab^{n-2} + b^{n-1})$$

泰勒：

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n + R_n(x)$$

$$R_n(x) = \begin{cases} \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1} \rightarrow \text{拉格朗日型余项} \\ o((x-x_0)^n) \rightarrow \text{佩亚诺型余项} \end{cases}$$

分段点左右处的导数要用定义求解。

$$x \rightarrow 0$$

$$x - \ln(1+x) \sim \frac{1}{2}x^2$$

$$a^x - 1 \sim x \ln a$$

$$\log_a(1+x) \sim \frac{x}{\ln a}$$

$$(1+x)^a - 1 \sim ax$$

$$\sqrt[n]{1+x} - 1 \sim \frac{1}{n}x$$

$$\tan x \sim \sin x \sim \frac{1}{2}x^3$$

$$\tan x + \sin x \sim 2x$$

$$\tan x - x \sim \frac{1}{3}x^3 \sim x - \arctan x$$

$$x - \sin x \sim \frac{1}{6}x^3 \sim \arcsin x - x$$

麦克劳林 ($x_0 = 0$)

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(n)}(0)}{n!}x^n + R_n(x)$$

$$R_n(x) = \begin{cases} \frac{f^{(n+1)}(\xi)}{(n+1)!}x^{n+1} \\ o(x^{n+1}) \end{cases}$$

主部是其低阶项，主部不是唯一。

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$$

$$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$$

$$(\arctan x)' = \frac{1}{1+x^2}$$

$$(\operatorname{arccot} x)' = -\frac{1}{1+x^2}$$

$$(\sec x)' = \sec x \tan x$$

$$(\csc x)' = -\csc x \cot x$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + o(x^n)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + \frac{(-1)^{n-1}x^{2n-1}}{(2n-1)!} + o(x^{2n})$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + \frac{(-1)^n x^{2n}}{(2n)!} + o(x^{2n+1})$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + \frac{(-1)^{n-1}x^n}{n} + o(x^n)$$

$$\frac{1}{1+x} = (1+x)^{-1} = 1 - x + x^2 - \dots + (-1)^n x^n + o(x^n)$$

$$(a^x)^{(n)} = a^x \ln^n a$$

$$(e^{ax})^{(n)} = a^n e^{ax}$$

$$(\sin x)^{(n)} = \sin(x + \frac{n\pi}{2})$$

$$(\cos x)^{(n)} = \cos(x + \frac{n\pi}{2})$$

$$(\sin kx)^{(n)} = k^n \sin(kx + \frac{n\pi}{2})$$

$$(\cos kx)^{(n)} = k^n \cos(kx + \frac{n\pi}{2})$$

$$(\ln x)^{(n)} = \frac{(-1)^{n-1}(n-1)!}{x^n}$$

$$\left(\frac{1}{1+x}\right)^{(n)} = \frac{(-1)^n n!}{(1+x)^{n+1}}$$

$$(1+x)^a = 1 + ax + \frac{a(a-1)}{2!}x^2 + \dots + \frac{a(a-1)\dots(a-n+1)}{n!}x^n + o(x^n)$$
 要考虑左右极限：
 分段点： $\frac{1}{x}$ 或 $\frac{1}{x-1}$...
 $\arctan \frac{1}{x}$
 $\frac{1}{1+e^x}$

$$(x^n)^{(n)} = n!$$
 莱布尼兹公式： $(u \cdot v)^{(n)} = \sum_{k=0}^n C_n^k u^{(k)} v^{(n-k)}$
 罗尔： $f(\xi) = 0$
 拉格朗日： $f(b) - f(a) = f'(\xi)(b-a)$
 柯西： $\frac{f(b)-f(a)}{F(b)-F(a)} = \frac{f'(\xi)}{F'(\xi)}$

$$\lim_{n \rightarrow \infty} \sqrt[n]{a^n + b^n + c^n + d^n} = \max\{a, b, c, d\}$$

夹逼： $\sqrt[n]{A^n} \leq \sqrt[n]{a^n + b^n + c^n + d^n} \leq \sqrt[n]{4A^n}$ 极值点： $f'(x) > 0$, 凹
 $A \leq \leq A \cdot \sqrt[n]{4}$ 极值点： $f'(x) < 0$, 凸

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$
 拐点可能在 $y'' = 0$ 处 或 y'' 不存在处

$$\int \sec x dx = \ln|\sec x + \tan x| + C$$

$$\int \csc x dx = \ln|\csc x - \cot x| + C$$

$$\tan \frac{x}{2} = \frac{1-\cos x}{\sin x} \quad \tan^2 x = \sec^2 x - 1$$

$$\cot^2 x = \csc^2 x - 1$$

$$\int \frac{1}{\sqrt{ax^2+bx+c}} dx \rightarrow \text{凑} \int \frac{1}{\sqrt{a^2+x^2}} dx \text{ 形式}$$

$$\int_a^b f(x)g(x)dx = f(\xi) \int_a^b g(x)dx$$

$$\int_a^b f(x)dx = f(\xi)(b-a)$$

积分估值:

若 $f(x)$ 有最大值 M , 最小值 m .

$$\therefore m(b-a) \leq \int_a^b f(x)dx \leq M(b-a)$$

换元必换限

定积分关于奇偶性的应用

点火公式:

$$\int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \cos^n x dx$$

↑ 区间长度

$$= \begin{cases} \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} & n \text{ 为偶} \\ \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{2}{3} & n \text{ 为大于1的奇数} \end{cases}$$

$$\text{柱壳: } V = \int -2\pi xy dx$$

关于齐次方程: 令 $u = \frac{y}{x}$, $y = ux$.

$$\frac{dy}{dx} = u + x \frac{du}{dx} \quad \text{代入原方程后分离变量}$$

$$\frac{dy}{dx} + P(x)y = Q(x)$$

① 当 $Q(x) \equiv 0$ 时, 分离变量

② 当 $Q(x) \neq 0$ 时,

$$y = \left(\int Q(x) e^{\int P(x) dx} dx + C \right) e^{-\int P(x) dx}$$

注意反函数的应用

伯努利: $\frac{dy}{dx} + P(x)y = Q(x)y^n$

令 $z = y^{1-n}$

$$\frac{dz}{dx} + (1-n)P(x)z = Q(1-n)Q(x)$$

$$z = \left(\int (1-n)Q(x) e^{\int (1-n)P(x) dx} dx + C \right) e^{-\int (1-n)P(x) dx}$$

$$= y^{1-n}$$

最后解出 y .

三. 变量降阶:

① $y^{(n)} = f(x)$

连续积分

② $y'' = f(x, y')$

令 $y' = p(x)$

$y'' = \frac{dp}{dx}$

③ $y'' = f(y, y')$

令 $y' = p(y)$

$y'' = p \cdot \frac{dp}{dy}$

<一> 高阶线性微分方程 + 多元函数微分.
 特解形式, 特解, 通解形式, 通解.
 非齐次通解 = 齐次通 + 非特.

对于齐次: $y'' + py' + qy = 0$.
 特征方程: $r^2 + pr + q = 0$.
 $\downarrow$
 $\begin{cases} \Delta = p^2 - 4q > 0, r_1 \neq r_2 \in \mathbb{R}, y = C_1 e^{r_1 x} + C_2 e^{r_2 x} \\ \Delta = 0, r_1 = r_2 \in \mathbb{R}, y = (C_1 + C_2 x) e^{r_1 x} \\ \Delta < 0, r_1 \neq r_2 \in \mathbb{C}, y = e^{\alpha x} [C_1 \cos \beta x + C_2 \sin \beta x] \end{cases}$
 $r_1 = \alpha + i\beta, r_2 = \alpha - i\beta$

对于非齐次: $y'' + py' + qy = f(x)$.
 ① $f(x) = e^{\lambda x} P_m(x)$. λ, r_1, r_2
 令 $y^* = x^k e^{\lambda x} Q_m(x)$. $k = \begin{cases} 0, & \lambda \text{ 不是根.} \\ 1, & \lambda \text{ 是单根} \\ 2, & \lambda \text{ 是重根.} \end{cases}$

② $f(x) = e^{\lambda x} [P_m(x) \cos \omega x + P_n(x) \sin \omega x]$.
 令 $y^* = x^k e^{\lambda x} [R_m(x) \cos \omega x + R_n(x) \sin \omega x]$.
 $k = \begin{cases} 0, & \lambda \pm i\omega \text{ 不是根} \\ 1, & \lambda \pm i\omega \text{ 是单根} \\ s, & \lambda \pm i\omega \text{ 为 } s \text{ 重根} \end{cases}$ $R_m^{(1)}, R_n^{(1)}$ 是 m 次多项式. $m = \max\{L, n\}$.

<二> 多元函数求导, 隐函数求导, 微分.
 $f_1, f_2, f_{12} = f_{21}$.
 1. 全微分: $dz = z_x dx + z_y dy$.
 2. 隐函数方程, 解方程.

<三> 方向导数 (在).
 $\frac{\partial f}{\partial l} = \frac{\partial f}{\partial x} \cos \alpha + \frac{\partial f}{\partial y} \cos \beta$. α, β 为方向角.
 $\vec{l} = (\cos \alpha, \cos \beta)$ 单位向量.
 方向导数的最大值为梯度的模.
 最小为梯度的负值.
 $|\text{grad} f(x, y)| = \sqrt{(\frac{\partial f}{\partial x})^2 + (\frac{\partial f}{\partial y})^2} = \max_{\alpha, \beta} \frac{\partial f}{\partial l}$.

$$\text{grad} f = \left\{ \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\}_m.$$

<四> 二重积分, 交换积分次序.
 1. 奇偶性运用.

$$d\sigma = dx dy = \rho d\rho d\theta.$$

$$2. \begin{cases} x = \rho \cos \theta \\ y = \rho \sin \theta. \end{cases}$$

3. 三重积分: 轮换对称性的运用.

$$\iiint_{\Omega} f(x, y, z) dV = \iiint_{\Omega} f(y, z, x) dV = \iiint_{\Omega} f(z, x, y) dV = \frac{1}{3} \iiint_{\Omega} [f(x, y, z) + f(y, z, x) + f(z, x, y)] dV.$$

① 柱坐标: $\begin{cases} x = \rho \cos \theta \\ y = \rho \sin \theta \\ z = z. \end{cases} dv = \rho d\rho d\theta dz.$

② 球坐标: $\begin{cases} x = r \sin \varphi \cos \theta \\ y = r \sin \varphi \sin \theta \\ z = r \cos \varphi \end{cases} dv = r^2 \sin \varphi dr d\varphi d\theta.$

在不能想出来或通过图知道 x, y, z 时可

通过计算.

$$\int_0^{\frac{\pi}{2}} \cos^n \theta d\theta = \int_0^{\frac{\pi}{2}} \sin^n \theta d\theta = \begin{cases} \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} & n \text{ 为偶正} \\ \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{3}{4}, n \text{ 为奇正} \end{cases}$$

4. 曲面面积: $A = \iint_D \frac{1}{\cos \gamma} d\sigma$.
 $z = f(x, y)$. $\cos \gamma = \frac{1}{\sqrt{1 + f_x^2 + f_y^2}}$

$F(x, y, z) = 0$ $\cos \gamma = \frac{F_z}{\sqrt{F_x^2 + F_y^2 + F_z^2}}$

<五> 曲线积分与路径无关简化曲线积分.

1. $x = \varphi(t), y = \psi(t)$. 对称性

$$\int_L f(x, y) d\sigma = \int_a^b f[\varphi(t), \psi(t)] \sqrt{\varphi'(t)^2 + \psi'(t)^2} dt.$$

$L: a \rightarrow b$.

2. 对全标的:

$$\int_L p dx + q dy = \int_a^b [p[\varphi(t), \psi(t)] \varphi'(t) + q[\varphi(t), \psi(t)] \psi'(t)] dt.$$

$L: a \rightarrow b$.

3. 两类曲线积分关系:

$$\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}$$

$$\vec{r} = (r(t), \varphi(t))$$

$$\text{反之 } \vec{r} = -1(r(t), \varphi(t))$$

方向余弦 α, β, γ

$$\int_L p dx + q dy = \int_L (p \cos \alpha + q \cos \beta) ds$$

4. 奇偶、轮换的应用

5. 格林公式: $\star$

$$\oint_L p dx + q dy = \iint_D \left(\frac{\partial q}{\partial x} - \frac{\partial p}{\partial y} \right) dx dy$$

当 $\frac{\partial p}{\partial y} = \frac{\partial q}{\partial x}$ 即积分与路径无关。 换解路径

<八> 高斯公式: 封闭曲面

$$\textcircled{1} \text{第一类曲面积分: } \iint_{\Sigma} f(x, y, z) dS = \iint_{D_{xy}} f(x, y, z(x, y)) \sqrt{1+z_x^2+z_y^2} dx dy$$

$$\sqrt{1+z_x^2+z_y^2} dx dy$$

$$\textcircled{2} \text{第二类: } \iint_{\Sigma} R(x, y, z) dx dy = \iint_{D_{xy}} R(x, y, z(x, y)) dx dy$$

注意正负号

$$\text{关系: } \iint_{\Sigma} p dy dz + q dz dx + R dx dy = \iint_{\Sigma} (p \cos \alpha + q \cos \beta + R \cos \gamma) dS$$

$\vec{n} = (-z_x, -z_y, 1)$ 单位化后取分量为方向余弦 $\cos \alpha, \cos \beta, \cos \gamma$

$$z = z(x, y) \Rightarrow z - z(x, y) = 0$$

$$\textcircled{3} \text{高斯公式: } \iiint_{\Sigma} p dy dz + q dz dx + R dx dy = \iiint_{\Omega} \left(\frac{\partial p}{\partial x} + \frac{\partial q}{\partial y} + \frac{\partial R}{\partial z} \right) dv = \iint_{\Sigma} (p \cos \alpha + q \cos \beta + R \cos \gamma) dS$$

<九> 收敛级数

$$\text{等比级数 } \sum_{n=0}^{\infty} a q^n = \begin{cases} |q| < 1, & \text{收敛} \\ |q| \geq 1, & \text{发散} \end{cases}$$

$$\text{调和级数 } \sum_{n=1}^{\infty} \frac{1}{n} \text{ 发散} \star$$

$$p\text{-级数 } \sum_{n=1}^{\infty} \frac{1}{n^p} \begin{cases} \text{收敛, } p > 1 \\ \text{发散, } p \leq 1 \end{cases}$$

1. 比较审敛法: 大收小收; 小发大发

$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0 \quad p > 0$$

$0 < l < +\infty$ 二者敛散性相同

$l = 0$ 二者收敛

$l = +\infty$ 二者发散

$$3. \text{比值审敛法: } \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \rho$$

$$\begin{cases} \text{收敛, } \rho < 1 \\ \text{发散, } \rho > 1 \end{cases} \quad \rho = 1 \text{ 失效}$$

$$4. \text{根值判别法: } \lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \rho$$

$$\begin{cases} \rho < 1, & \text{收敛} \\ \rho > 1, & \text{发散} \end{cases}$$

交错级数 $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^p}$, $p > 0$ 时收敛

5. $\sum_{n=1}^{\infty} |u_n|$ 收敛 $\Rightarrow \sum_{n=1}^{\infty} u_n$ 收敛 (绝对)

$\sum_{n=1}^{\infty} |u_n|$ 发散, 而 $\sum_{n=1}^{\infty} u_n$ 收敛, 则 $\sum_{n=1}^{\infty} u_n$ 条件收敛

<八> 幂级数 $\star$

$$1. \sum_{n=0}^{\infty} (-x)^n = \frac{1}{1-x}, |x| < 1$$

$$\sum_{n=1}^{\infty} (x^n)' = \left(\frac{x}{1-x} \right)' \quad \sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad \sum_{n=1}^{\infty} x^n = \frac{1}{1-x} - 1 = \frac{x}{1-x}$$

2. 收敛区间与收敛域不同

收敛域要讨论区间端点处的收敛性

收敛性

$$\text{对于 } \sum_{n=0}^{\infty} a_n x^n \quad (a_n \neq 0)$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1} x^{n+1}}{a_n x^n} \right| = \left| \frac{a_{n+1}}{a_n} \right| |x| = \rho |x|$$

$$\textcircled{1} \rho \neq 0 \text{ 时, } |x| < \frac{1}{\rho} \text{ 收敛, } |x| > \frac{1}{\rho} \text{ 发散. } R = \frac{1}{\rho}$$

$$\textcircled{2} \rho = 0 \text{ 时, } R = +\infty$$

$$\textcircled{3} \rho = +\infty \text{ 时, } R = 0$$

3. 求和函数 $S(x)$ { 求导, 先求收敛域, 积分 }

4. 函数展成幂级数. 记得写范围.

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, x \in \mathbb{R}.$$

$$\sinh x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, x \in \mathbb{R}$$

$$\cosh x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, x \in \mathbb{R}.$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} (-1)^n x^n, x \in (-1, 1).$$

$$\ln(1+x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}, x \in (-1, 1].$$

5. 傅立叶:

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx.$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx). \quad x \in _$$

收敛定理:

x 是连续点时, 级数收敛于 $f(x)$.

x 是间断点时, 收敛于 $\frac{f(x) + f(x^+)}{2}$.

对 $f(x)$ 进行 $\begin{cases} \text{奇延拓, 只有正弦.} \\ \text{偶延拓, 只有余弦.} \end{cases}$

周期延拓.

一般周期 $2L$ 的傅里叶: 同上.

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx.$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx.$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L}) \quad x \in _$$

10 <九>

1. 对于 $w^n = z$. 将 z 改写成三角式.

$$w = \sqrt[n]{z} = r^{\frac{1}{n}} \left(\cos \frac{\theta + 2k\pi}{n} + i \sin \frac{\theta + 2k\pi}{n} \right).$$

$$z^a = e^{a \ln z} = e^a$$

2. C-R 方程: $f(x) = u(x, y) + i v(x, y)$.

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases} \quad \text{成立, 则在该点处可导,}$$

3. 不解析的点为奇点.

4. $f(x) = u + i v$ 在 D 内解析 $\Leftrightarrow u_x, u_y, v_x, v_y$ 连续 要说明.

且满足 C-R 方程.

5. $\ln z = \ln|z| + i(\arg z + 2k\pi)$.
复变函数积分. $z^a = e^{a \ln z} = e^{a(\ln|z| + i(\arg z + 2k\pi))}$

$$\textcircled{1} \int_C f(z) dz = \int_C (u + i v) (dx + i dy).$$

$$\textcircled{2} z = z(t) = x(t) + i y(t), \quad t: a \rightarrow b$$

$$\int_C f(z) dz = \int_a^b f(z(t)) z'(t) dt.$$

柯西定理: $f(z)$ 在 D 内解析, C 为闭曲线

$$\oint_C f(z) dz = 0.$$

$$\oint_C \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0).$$

$$\oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz = \frac{2\pi i}{n!} f^{(n)}(z_0).$$